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Enhancement of OCT signal interpretability in deep penetration laser welding of aluminum

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ABSTRACT

In laser deep penetration welding a constant weld depth is an essential quality measure that is usually obtained by in-line process monitoring techniques such as optical coherence tomography (OCT). However, due to the lack of referential information about the keyhole shape, the extraction of meaningful information about the keyhole from the OCT signal is limited to the keyhole depth which is acquired by means of statistical analyses. In this research on-line X-ray-recordings alongside OCT measurements were conducted during the welding of pure aluminum (EN AW-1050A) in the Rosenthal regime to enable a more profound interpretation of the OCT signal. From the X-ray recordings of the process zone, 3D keyhole shapes were reconstructed and meaningfully reduced to a set of archetypical keyhole geometries by means of a hierarchical agglomerative cluster analysis. It was found that these archetypical keyhole shapes are not unique for individual experiments but represent prevalent keyhole states that largely remain valid even across experiments conducted with different process parameters. Investigating the interconnections between geometric archetypical keyhole features and features of their corresponding OCT data point distributions revealed significant correlations beyond the mere keyhole depth raising the potential of extending the applications of OCT in laser beam welding processes.

Introduction

The quality of weld seams obtained from laser deep penetration welding is primarily determined by the process dynamics. Turbulent dynamics can lead to critical process defects like blow-outs, spatter as well as the formation of pores. Thus, the surveillance of the process dynamics raises the possibility of an in-line detection of process defects given a sufficient correlation between the monitoring signal and the process defect. In laser deep penetration welding the process dynamics is primarily governed by the keyhole which constitutes the center region of laser absorption and is formed by the laser induced ablation pressure acting on the melt pool surface (Svenungsson et al., 2015). To gather information about the keyhole dynamics many experimental approaches have been investigated throughout the past including the measurement of electromagnetic emissions through the keyhole opening (Engelhardt et al., 2015), airborne sound emissions (Gu and Duley, 1996; Weiss et al., 2024) as well as methods for the characterization of geometric features of the keyhole including their temporal evolution. The latter mentioned methods for geometric characterization are considered the most promising as they yield insight into the system of the process zone beyond mere correlations between the monitoring signal and properties of the resulting weld seam. For this purpose, X-ray and synchrotron measurements of the process zone as well as optical coherence tomography (OCT) have emerged as the most insightful and applicable methods. While in-line monitoring of the process zone by means of high-speed X-ray or synchrotron recordings yields a time-resolved series of two-dimensional images of the process zone revealing detailed information about the keyhole shape and its temporal evolution it comes at high experimental costs and despite its high information density it is not suited for implementation in production facilities. OCT on the other hand raises a simple distance measurement that, when directed into the keyhole opening coaxially to the processing laser beam, yields information about the distribution of reflective surfaces inside the keyhole. As the keyhole ground constitutes the most prominent geometric keyhole feature detected by the probe beam, monitoring the penetration depth of the laser beam is the most widely spread application of this

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measurement technique. However, as previously highlighted by Pordzik et al. (2023), the OCT raw signal contains information beyond the keyhole depth. From this insight the question arises whether features of the OCT signal that are often discarded as noise can be attributed to actual keyhole phenomena and thus raise the potential of an enhanced interpretability of the OCT signal beyond the mere evaluation of the keyhole depth and its temporal fluctuations.

X-ray recordings of the process zone are most commonly applied technique to acquire a time-resolved lateral projection of the process zone from which the melt pool and the keyhole can bis distinguished due to their difference in density and thus different absorption properties with respect to the X-ray beam. The partial extinction of the X-ray radiation according to the Beer-Lambert law manifests in different brightness of certain regions on the resulting image (Boley et al., 2019). Due to the small difference in density between the solid and the liquid material phase, the solid-liquid interface is less pronounced. The keyhole on the other hand, being filled with surrounding gas and metallic vapor exhibits a vastly different density resulting in a sharp contrast between the keyhole and its surroundings. Boley et al. performed a 3D reconstruction of the melt pool and keyhole shape during deep penetration laser welding of aluminum and iron where they used tungsten tracers to enhance the contrast between the solid and liquid phase (Boley et al., 2013). Time-resolved 3D reconstructions of the keyhole on the basis of X-ray recordings were retrieved be Fetzer et al. and combined with ray-tracing simulations to obtain the local absorption of the processing laser beam on the keyhole walls (Fetzer et al., 2021). Reinheimer et al. investigated the influence of the keyhole geometry on the quality of the resulting weld seams at high traveling velocities (Reinheimer et al., 2022). A stabilizing influence of the stronger inclination of the keyhole front wall at higher traveling velocities for welding of copper was identified by Kaufmann et al. (2023). Furthermore, significant differences in the keyhole shape and its sensitivity to process parameters between steel and copper at different laser wavelengths were observed (Kaufmann et al., 2024). Schricker et al. were able to characterize the interaction between the keyhole and the molten pool during copper welding using high-resolution synchrotron images of the process zone (Schricker et al., 2022). Kawahito et al. concluded from X-ray measurements of the process zone that up to a velocity of 100 mm/s, a large part of the laser power is absorbed at the bottom of the keyhole and that the spot only shifts toward the front wall at higher speeds (Kawahito et al., 2011). The time evolution of keyhole shapes during the welding of titanium was investigated based on high-speed synchrotron recordings of the process zone by Pyeon et al. who employed feature extraction techniques to describe the 2D keyhole shapes (Pyeon et al., 2021). Thus, they were able to identify correlations between the process parameters and the resulting geometric keyhole features.

Optical coherence tomography (OCT) is a well-established method for probing the keyhole depth in-line by performing a distance measurement via a probe beam that is guided into the keyhole coaxially to the processing laser beam (Bautze and Kogel-Hollacher, 2014) . A scanning OCT measuring beam also enables the mapping of a three-dimensional profile of the keyhole as a temporal average (Sokolov et al., 2020). This was implemented by Sokolov et al. to control the welding depth based on the keyhole depth and other characteristics of the OCT signal related to the keyhole shape. The influence of the keyhole shape on the OCT signal was investigated by Werner et al., who found that the measurement angle between the OCT beam and the surface had a significant influence on the weld depth measurement (Werner et al., 2022). Fleming et al. validated the OCT measurement signal in terms of its actual significance with regard to the welding depth by performing synchrotron imaging of the process zone synchronously with the OCT measurements (Fleming et al., 2023). By directly comparing the OCT signal with the keyhole geometry, it was possible to interpret the characteristics of the OCT signal in a meaningful way, with a high degree of agreement between the depth information provided by both methods when welding an aluminum alloy with a vigorously fluctuating keyhole shape. In most cases, especially with process parameters that result in turbulent keyhole dynamics, statistical filtering of the OCT signal is required in order to extract a plausible value for the weld penetration depth. Mittelstädt et al. presented an approach for a histographic evaluation of the measurement signal, in which the last local maximum of the histogram was associated with the weld penetration depth (Mittelstädt et al., 2018). Pordzik et al. demonstrated that the significantly reduced keyhole dynamics lead to a considerable improvement in the OCT signal by comparing welds in aluminum and steel in a vacuum and at atmospheric ambient pressure (Pordzik et al., 2022). Furthermore, it was shown that taking into account an extended part of the OCT frequency spectrum leads to a higher information density and thus to a more robust detection of the keyhole depth (Pordzik et al., 2023). Another method for extracting the weld penetration depth from the raw OCT signal was developed by Boley et al., which allows a parameter-independent analysis of the weld penetration depth by applying a noise filter (Boley et al., 2019). It was demonstrated by Pordzik and Seefeld that OCT can be utilized to detect asymmetric keyhole deformations in the lateral direction and employ this information using artificial neural networks to distinguish between the directions of weld path deviations during the welding of hidden T-joints (Pordzik and Seefeld, 2024). Besides investigations of the keyhole depth fluctuations and their implications on the process stability and weld seam quality OCT has been also used to investigate absorption properties of the keyhole. For this purpose, Allen et al. combined OCT measurements with measurements of the scattered laser power by an integrating sphere to find correlations between the overall weld penetration depth and the keyhole absorptance regarding the laser beam (Allen et al., 2020).

This research aims at enhancing the interpretability of the OCT signal by linking characteristics of the OCT signal to geometric features of the underlying keyhole shapes. Therefore, the total set of keyhole shapes occurring during a single welding process is systematically reduced by a hierarchical agglomerative clustering approach sorting the keyholes based on their similarities into keyhole clusters each representing an archetype of keyhole configuration at a certain level of abstraction. The level of abstraction is determined by the chosen target number of clusters N. With respect to the generalizability of keyhole clusters it is investigated whether different repetitions of a welding process yield the same or similar clusters as well as whether similarities between the archetypical keyhole shapes between welding processes with different laser powers can be identified. A scheme of the process zone during laser deep penetration welding yielding the OCT features and keyhole geometries is shown in Fig. 1.

Based on the geometric clustering of the keyhole shapes, the histographic distributions of OCT data points belonging to each archetype are characterized and mapped onto the geometric features characterizing each keyhole cluster. This procedure allows for the identification of potential correlations between the OCT signal and the underlying keyhole shape, thus raising the potential of obtaining information about the keyhole shape other than the mere keyhole depth solely based on OCT measurements. Significant correlations between both feature spaces would considerably improve the diagnostic potential of OCT for in-line detection or even prognosis of process errors during deep penetration laser welding. A flow chart of the applied analysis methods and key aspects of the presented research can be found in Fig. 2.

Experimental setup and methods

Experimental setup

To investigate the correlations between characteristics of the OCT signal with features of the actual keyhole shape bead on plate weldings were carried out in pure aluminum (EN AW-1050A) at a constant traveling velocity of 3000 mm/min. The weld regime at the given velocity is

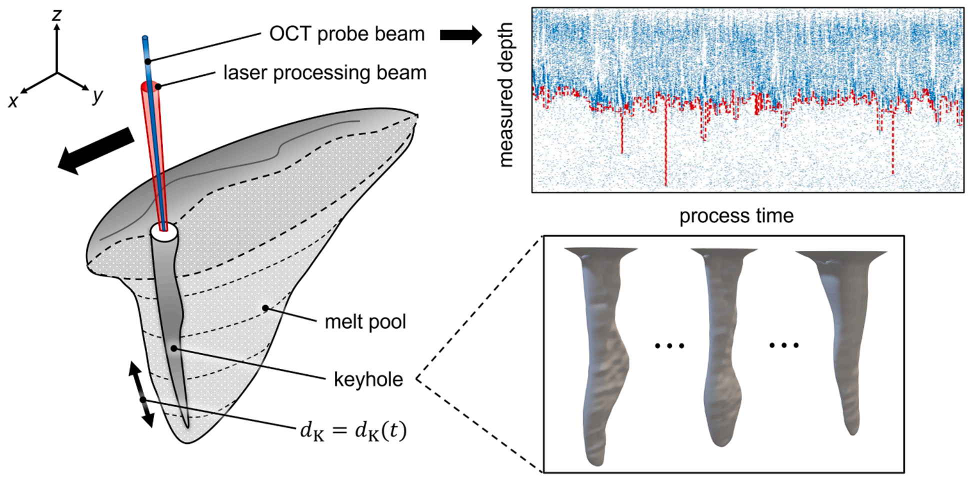

Fig. 1. Schematic of a process zone during a laser deep penetration welding process accompanied by OCT measurements with exemplary data obtained from these OCT measurements and X-ray based 3D keyhole reconstructions.

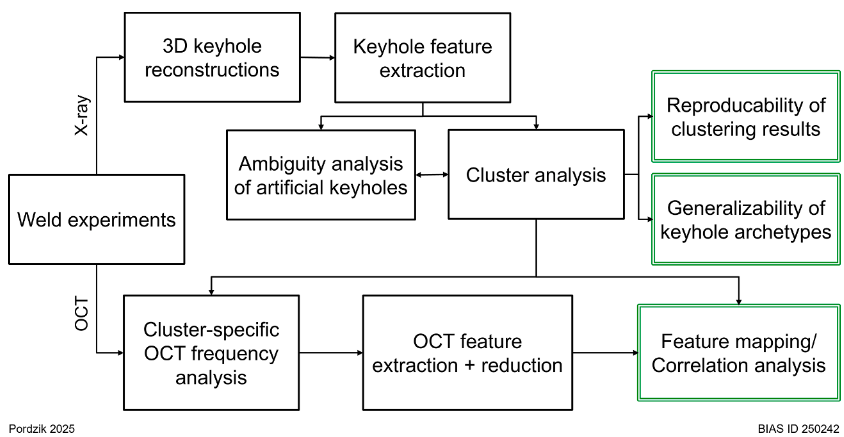

Fig. 2. Graphical structure of the conducted analyses and key aspects of the presented research.

also referred to as Rosenthal regime and is characterized by a rather straight keyhole axis as well as by a weak melt pool convection making heat conduction the dominant heat transport mechanism inside the process zone (Fabbro, 2010). Three different laser powers ranging from 4 kW to 6 kW were applied resulting in three different weld penetration depths each yielding different geometric features of the resulting ensemble of keyhole shapes. The length of the weld seams was 80 mm resulting in a process duration of approximately 1.6 s per weld seam. The weld samples had a length of 100 mm, a height of 35 mm and a width of 5 mm. The rather small width of the weld sample potentially resulting in an increased heat accumulation is a requirement for sufficient X-ray transmission. The weld sample was placed in a clamping device mounted on a motorized one-dimensional axis allowing the sample to be moved at the desired traveling velocity. Moving the sample relative to the processing head and the X-ray setup the process zone remained stationary with respect to the measuring devices. The disk laser TruDisk 16002 (Trumpf SE + Co. Kg.) was used as laser beam source operating at a wavelength of 1030 nm with a beam parameter product of 4 mm · mrad at a maximum power of 16 kW. The laser beam was directed to the processing head via a fiber optic cable with a core diameter of 200 μm. The processing head of the type YW52 (Precitec GmbH & Co. Kg.) had a magnification ratio of 3:2 and was equipped with an OCT module IDM (Precitec GmbH & Co. Kg.) which allowed the OCT probe beam to be directed into the keyhole coaxially to the beam of the processing laser. The resulting laser beam had a Rayleigh length of 2.52 mm and a nominal focal diameter of 300 μm. The focal plane of the laser beam was shifted beneath the surface of the weld sample resulting in a spot diameter at the sample surface of 600 μm and a converging laser beam caustic. A leading beam inclination of −5° with respect to the surface normal of the weld sample was applied, tilting the laser beam towards the travel direction. The OCT was operated at a central wavelength of 1550 nm with 35 nm full width at half maximum. The spot diameter of the probe beam was 70 μm while the sampling rate of the system was 70 kHz. Test weldments were carried out to accurately adjust the OCT probe beam towards the central axis of the keyhole, thus maximizing the quality of the measured OCT signal. A schematic depiction of the experimental setup including the applied process parameters is presented in Fig. 3.

A setup consisting of an X-ray tube, a scintillator and a high-speed camera was employed for the on-line X-ray recordings of the process zone during the laser welding process. The X-ray tube FXE-225.48 (Comet Xylon GmbH) served as the source of radiation providing a minimal spot diameter of 6 μm operating an acceleration voltage of 150 kV. The imaging was performed using a scintillator by the manufacturer Hamamatsu Photonics K.K. providing a resolution of 728 × 728 px. The X-ray images on the scintillator were recorded by the high-speed camera Os8 (Imaging Solutions GmbH) at a framerate of 5000 fps with a resolution of 768 × 768 px. In combination the setup provided an effective spatial resolution of 15 μm/px. At a given framerate of the X-ray setup of 5000 fps a total of 4000 to 4500 X-ray images were captured per weld experiment discarding measurement from the start and the end of the process which are not considered to be representative for the

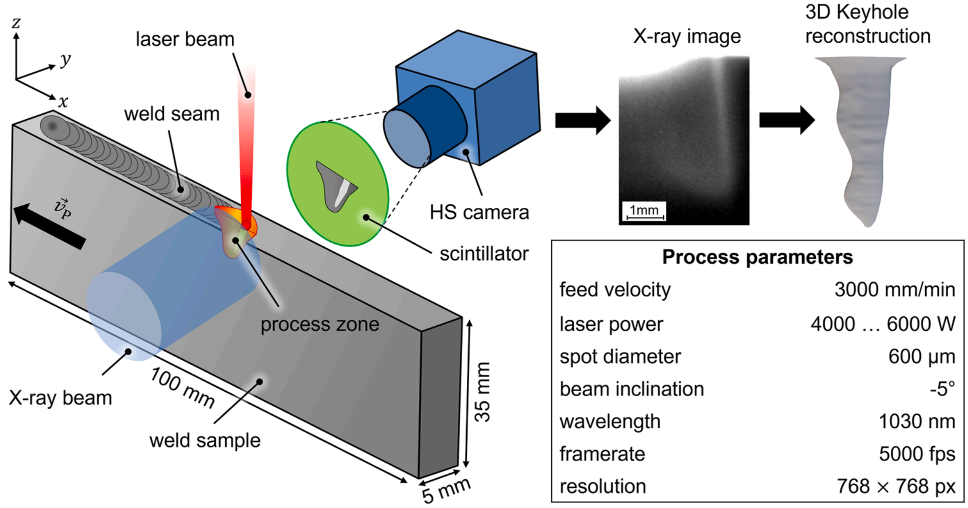

Fig. 3. Schematic of the experimental setup including the applied process parameters.

undisturbed process.

Methods

Reconstruction of keyhole geometries

The recorded X-ray videos were post-processed using Kalman filtering to enhance image contrast and reduce noise. To further improve the image quality, a flat-field correction was applied in order to compensate for the inhomogeneous spatial intensity distribution of the incident X-rays. A comprehensive description of the employed X-ray system and the corresponding image processing methods can be found in Boley et al. (2019) and Abt et al. (2011). The absorption of X-rays is governed by their wavelength, the density and atomic number of the specimen, and the path length through the irradiated material. When the X-rays pass through regions of the specimen with different path lengths—caused, for instance, by localized vaporization of the material under laser irradiation—the radiation experiences varying degrees of attenuation. In areas with shorter path lengths, corresponding to vapor-filled capillaries, the attenuation is reduced, leading to stronger scintillator fluorescence and an increase in the gray value of the camera image (Boley et al., 2019). In contrast, regions with longer path lengths, where the beam traverses liquid or solid material, exhibit lower gray values. By exploiting the relationship between X-ray attenuation and the recorded gray values, the transverse width and thus the three-dimensional geometry of the keyhole can be reconstructed. This reconstruction is based on Beer’s law, assuming a direct correlation between the measured intensity reduction and the effective material thickness along the X-ray path (Fetzer).

The X-ray imaging of the process zone was conducted in the plane defined by the laser beam axis and the traveling direction. The recorded grayscale images therefore represent line-of-sight projections of the local density field onto this plane. Consequently, depth information associated with asymmetric fluctuations perpendicular to the projection plane is inherently lost. Since the experimental configuration and boundary conditions establish this plane as a plane of symmetry, a symmetric process-zone geometry with respect to the projection plane was assumed for the reconstruction of the depth information.

Analysis of reconstructed 3D keyhole geometries

The 3D reconstructed geometries of the keyhole shapes derived from the X-ray images were analyzed regarding a set of geometric features so that each geometry could be characterized by a set of scalar values. Initially the following set of criteria was employed for the keyhole analysis:

- Depth d_K
- Volume V_K
- Surface area S_K
- Surface area / volume
- Keyhole front inclination α
- Bulbousness (absolute) B
- Bulbousness (normalized) $\widehat{B}$
- Aspect ratio (depth / length) κ_l
- Aspect ratio (depth / width) κ_w
- Aspect ratio (depth / circumference) κ_c

The analysis was performed by a fast custom-built algorithm implemented in C++. Each STL file containing a certain keyhole geometry is imported and sliced by a fixed number of equidistant planes that are oriented perpendicular to a specified analysis axis, resulting in N cross-sections of the keyhole (see Fig. 4a)). In this case, the laser beam axis was chosen as the analysis axis and for each geometry a total of 1000 slices were acquired. By evaluating the cross-sectional area as well as the circumference of each slice approximate values for the volume, the surface area and the bulbousness could be derived as shown in Fig. 4b) and c).

The front angle was obtained from the inner 80 % of the keyhole depth to exclude geometric deviations in the vicinity of the keyhole opening and the bottom from the analysis. The keyhole depth is defined by the STL node with the smallest z-coordinate while the aspect ratio is calculated as the ratio between the keyhole depth and the mean radius of the keyhole opening. The feature described as bulbousness was defined to serve as a measure for the constricted keyhole volume below a certain depth. Its main purpose lies in the investigation of the effect of possible multiple reflections inside the keyhole on the OCT signal. It is defined as the quotient between the actual volume beneath a certain depth and the extruded volume based on the cross-section at the regarding depth position.

$$b(h) := \frac{V(h)}{\widetilde{V}(h)} - 1 \tag{1}$$

$$B(h) = \int_0^h b(h')\,dh' \approx \sum_{i=1}^{n} b\left(\underbrace{h_i}_{=i\cdot\Delta h}\right)\cdot\Delta h \tag{2}$$

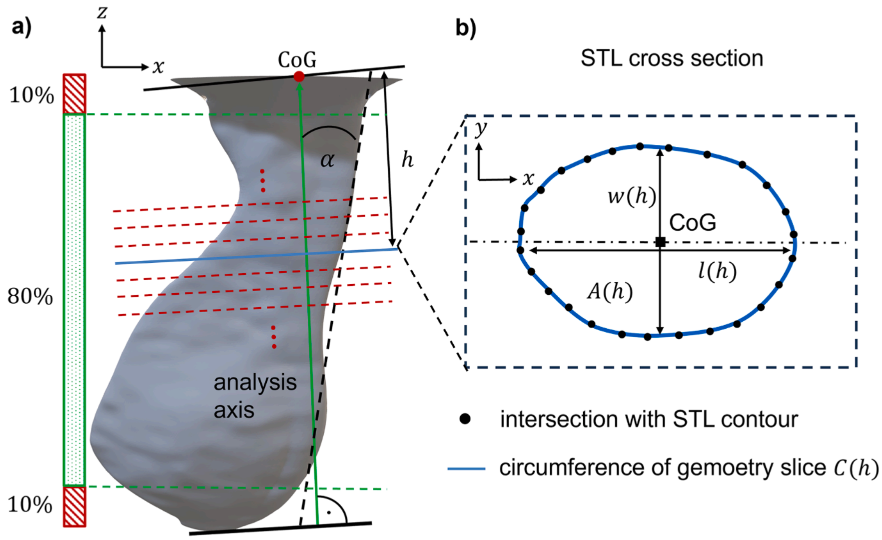

Fig. 4. Schematic of the feature extraction from 3D keyhole reconstructions: a) Slicing of the STL keyhole geometries; b) Geometric analysis of the resulting cross-sections; c) Graphical explanation of the bulbousness criterion.

$$\widehat{B}(h) = \frac{B(h)}{h} = \frac{\int_0^h b(h')\, dh'}{h} \approx \frac{1}{n}\sum_{i=1}^{n} b(h_i) \tag{3}$$

In Eqs. (1)–(3) the letter b denotes the local bulbousness at a given height h as illustrated in Fig. 4c). B denotes the total bulbousness at a certain height defined as the height integral of the local bulbousness $b(h)$ from the keyhole bottom up to the targeted height of evaluation. Finally, the normalized bulbousness $\widehat{B}$ is defined as the total bulbousness divided by the height of evaluation. Analyzing each reconstructed keyhole geometry of the weld seam, each keyhole configuration can be represented by a N-dimensional vector or as a point in the N-dimensional feature space respectively. This kind of representation of the keyhole shape configurations allows for the application of data analysis methods such as clustering algorithms to find underlying structures in the distribution of geometric features.

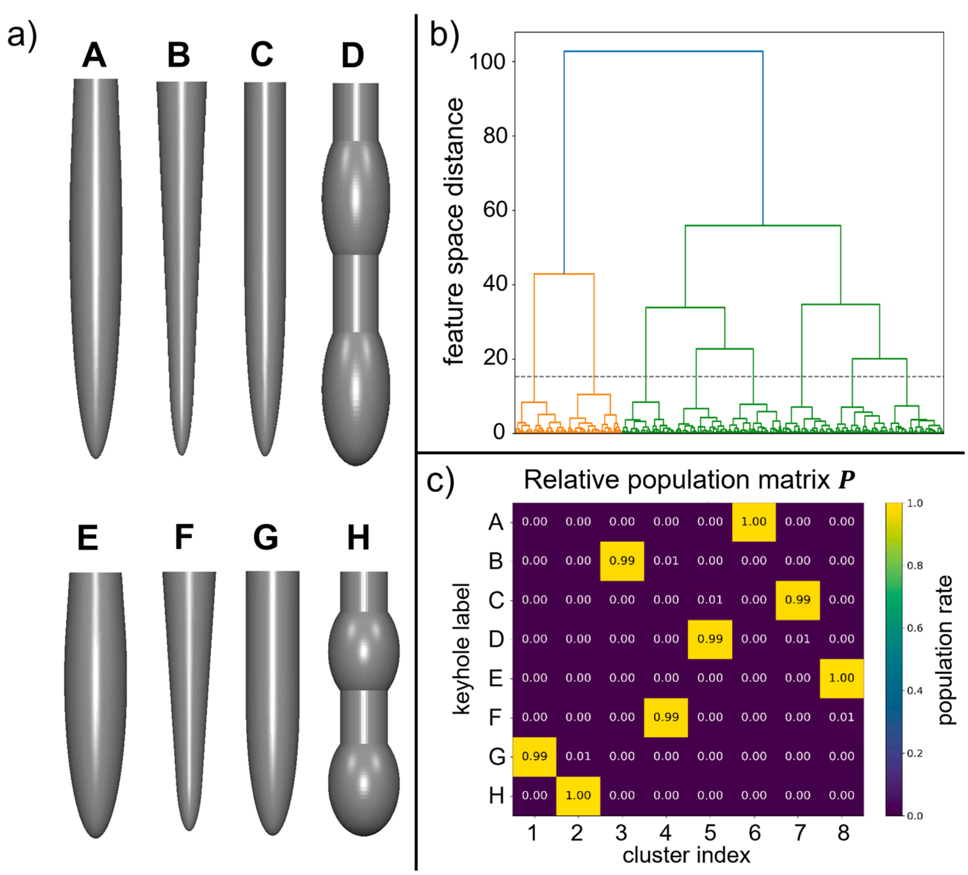

Fig. 5. Qualification of the proposed geometric keyhole features by an ambiguity analysis showing the labeled model geometries (a) along with the resulting clustering dendrogram (b) and the relative population matrix (c).

Hierarchical agglomerative clustering

To search for an underlying structure in the distribution of geometric features that were assigned to the reconstructed keyhole shapes the approach of hierarchical agglomerative clustering (HAC) was adopted. Hierarchical clustering is a method to identify data point accumulations within a given data set based on a metric of similarity between clusters and their constituents. Those similarities are defined as distances inside the feature space which are calculated based on a selected metric and so-called linkage between clusters.

Agglomerative clustering describes a bottom-up approach so that in the first clustering instance every data point constitutes a cluster on its own. In each consecutive instance the persisting clusters are merged pairwise based on the distance between them so that clusters which are most similar to one another become a new cluster on the next hierarchy level. In the last step only a single super cluster containing all data points remains. The clustering results are usually visualized in a dendrogram as shown in Fig. 5b). The horizontal lines indicate the merging of a new cluster whereas the similarity between both subclusters is quantified by the position of the horizontal line on the y-axis. The vertical lines represent the distances between a parent cluster and its children based on the applied metrics and linkage. Based on a distance threshold a subset of N clusters can be extracted. On the one hand, by increasing the distance threshold, the number of resulting clusters decreases while the similarity of data points inside each cluster decreases. On the other hand, decreasing the distance threshold results in a greater number of clusters with an increasing similarity between the members of each cluster.

The analyses were performed using the HAC algorithm provided by the Python library *Scikit-learn* (version 1.5.1) (Pedregosa et al., 2011). For the presented analyses, the keyhole representations in the geometry feature space were standardized feature-wise by subtracting the means and normalizing the features with respect to their respective standard deviations. This is necessary to treat all features as equal as otherwise features exhibiting different scales would contribute differently to the distance metrics and thus prioritize the importance of larger scaled features. Furthermore, outliers were defined as data points outside the 3σ-environment and excluded from further analyses. For the distance calculation in the feature space a Euclidean metrics was employed while the linkage between data sets was calculated via the so-called ward-linkage. Ward-linkage selects pairs of clusters to be merged by minimizing the gain in cluster variance and is a commonly used method. Thus, it is not to be confused with the Euclidean distance between the cluster centers. A target value of $N = 10$ clusters was set for the analyses of the experimentally obtained keyholes, and the required distance threshold was calculated accordingly. From the analyses the cluster populations, the clusters centers of gravity in feature space along with their feature-specific dispersions as well as the distance matrices indicating the similarity between the different clusters were retrieved.

$$a_j^{(\eta)} = 1 - \left\{ \max\left(\boldsymbol{p}_j\right) - \left[\frac{1}{N-1}\left(\sum_{i=1}^{N} p_{ij}^{\frac{1}{\eta}} - \max\left(\boldsymbol{p}_j\right)^{\frac{1}{\eta}}\right)\right]^{\eta} \right\},\ a_j^{(\eta)} \in [0,\ 1] \text{ and } \eta \in \mathbb{N}^{+} \tag{5}$$

In order to compare the results of cluster analyses between different welds, it is necessary to identify equivalences between clusters derived from different data sets. This is achieved by applying the so-called Hungarian algorithm to the distance matrix of the two cluster sets. This method ensures that each cluster of the first data set is exclusively paired with a single cluster from the second one while minimizing the average pairwise distance. As the global average distance between clusters is minimized, not all pairs necessarily yield the minimum possible distance. The less the distance between paired clusters, the higher the likelihood of being equivalent to one another. By averaging all pairwise distances between the different data sets a single scalar value is obtained that is associated with their overall cluster accordance.

Qualification of geometric feature criteria

The catalogue and the number of potential criteria for the characterization of keyhole geometries is completely arbitrary and a matter of choice for the analyst. Therefore, it is necessary to qualify and validate the selected criteria beforehand by applying them to artificially created model geometries. Thus, it is possible to validate their suitability and effectiveness for clustering the experimentally obtained keyhole geometries.

For this purpose, a set of four different cavity models was created which is assumed to reflect fundamental geometric properties of typical keyholes (see the cavity models A-D in Fig. 5a)). By scaling those geometries in the direction of their rotational axis from the four model geometries two depth cohorts were created resulting in a total of eight generic cavity models. Each model was then replicated 500 times while each copy was distorted in all cartesian coordinates with distortion factors that were generated randomly based on a gaussian probability distribution. Thus, a variety of different model geometries totaling 4000 geometries were created while each model geometry carried the label of the model type it was derived from. The widths of the gaussian distributions for the distortion factors were chosen such that an overlap between different model types regarding certain geometric features was ensured.

The generated qualification set of model geometries was then subjected to the geometry analysis as described in "Analysis of reconstructed 3D keyhole geometries" and then iteratively clustered hierarchically each time applying a different combination of feature criteria. The distance threshold for the clustering was chosen such that it would result in eight clusters or the same number of clusters as the number of different model types respectively. Finally, an ambiguity criterion was applied to the clustering results that was designed to quantify how well the applied cluster criteria performed in distinguishing between the model types identified by their respective label.

Let $\boldsymbol{P}$ be an $N \times M$ matrix that contains the relative population $p_{i,j}$ of a cluster i by elements of the class j where N denotes the number of model classes and M stands for the number of clusters. In the special case of analyzing the ambiguity of the clustering analysis regarding the data point labels the number of clusters is equal to the number of model classes $N = M$.

$$\boldsymbol{P} = \begin{bmatrix} p_{1,1} & \cdots & p_{N,\,1} \\ \vdots & \ddots & \vdots \\ p_{1,M} & \cdots & p_{N,M} \end{bmatrix} \text{with} \sum_{i=1}^{N} p_{ij} = 1 \tag{4}$$

$$A^{(\eta)} = \frac{1}{M}\sum_{j=1}^{M} a_j^{(\eta)}, A^{(\eta)} \in [0,\ 1] \tag{6}$$

In the Eqs. (5) and (6), a_j denotes the cluster-specific ambiguity that yields a value between 0 and 1 depending on the distribution of the class members on the different clusters. Hereby, 0 indicates the case of perfect classification where all class members are collected into an individual cluster whereas 1 indicates complete ambiguity referring to a state where the class members of class *j* are scattered homogenously across all clusters. The total ambiguity of the distribution is then denoted by A and is defined as the mean value of all cluster-specific ambiguities. The ideal outcome of such an analysis would be the case when all members of each class are assigned to an individual cluster resulting in $A = 0$. More concisely, the ambiguity criterion indicates how strongly the cluster, being mostly populated by members of a certain class, dominates over the mean population of all remaining clusters. The parameter η determines how strongly the ambiguity accounts for the scattering of class members across different clusters. The more broadly the class members are scattered the higher the ambiguity becomes. In this case the parameter was selected as $\eta = 2$.

By minimizing the ambiguity value from the initially proposed set of geometric features a subset was identified which showed the best performance in separating the distorted model geometries according to their labels resulting in a total ambiguity of $A = 0.0072$. The corresponding dendrogram and the population matrix are presented in Fig. 5b) and c). From the initially proposed keyhole features (see Analysis of reconstructed 3D keyhole geometries) the following were identified as the features with the least ambiguity regarding the clustering of labeled artificially created cavity models and employed for the geometric characterization of the experimentally obtained keyhole shapes:

- Depth d_K
- Surface area S_K
- Keyhole front inclinations α
- Bulbousness (normalized) $\widehat{B}$
- Aspect ratio (depth / circumference) κ_c

Histographic analysis of the OCT signal

The OCT data point distribution was analyzed according to Mittelstädt et al. using a histographic analysis (Mittelstädt et al., 2018). Hereby, the data points are sorted into depth bins with a bin width of 180 µm resulting in a frequency distribution of depth values occurring in the OCT raw signal. The bin width influences the level of detail of the resulting frequency distribution and thus determines the number of accessible features. A bin width of 180 µm was found to provide a reasonable level of detail only preserving characteristic features which are not attributed to statistical fluctuations of the underlying data point sample. As the cluster populations may differ and therefore cover different time intervals it is necessary to normalize the resulting histograms with respect to the total number of data points to ensure comparability. The histogram exhibits certain features which can potentially be assigned to geometric properties of its corresponding keyhole archetype. In an analogous manner to the characterization of the reconstructed keyhole geometries the OCT depth histograms are characterized by a set of criteria describing the most important features of the distributions, which are shown in Fig. 6.

The OCT measurements and the reconstructed keyhole geometries have a fixed temporal relation so that it is possible to collect all OCT data points belonging to a certain keyhole archetype and evaluate them cluster-wise. Thus, the cluster-specific histographic OCT frequency distributions, in the following referred to as OFD, can be obtained and analyzed regarding the introduced histogram features. Fig. 6 illustrates how the OCT data points belonging to a certain keyhole archetype are

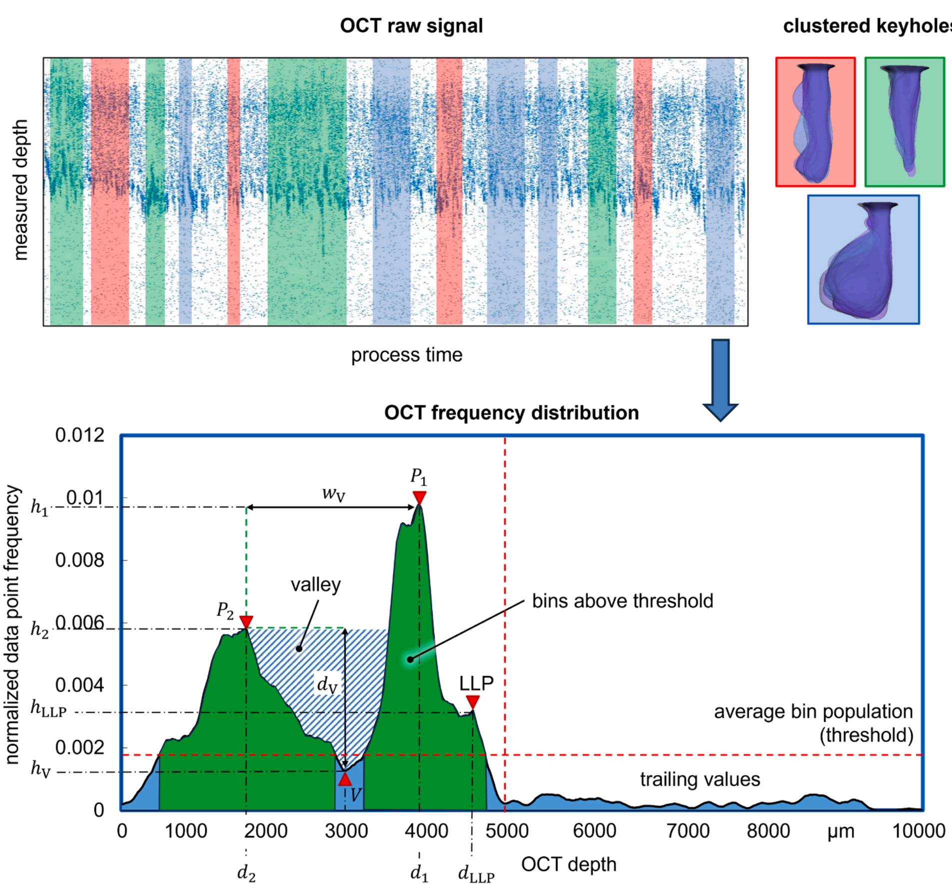

Fig. 6. Schematic of the OCT data collection based on the underlying keyhole archetypes (top) and the feature extraction from the resulting frequency distributions (bottom).

collected and analyzed histographically.

To characterize the OCT histograms an initial set of 22 features was applied which are mainly derived from the basic features highlighted in Fig. 6. The histograms are segmented into the main area on the left and the trailing area on the right. The main area is characterized by bin populations exceeding a significance threshold defined as the average bin population of the histogram. From this area the most prominent peaks are identified with emphasis on the most and second most prominent as well as the last local peak which is associated with the keyhole depth (Pordzik et al., 2022). Furthermore, the valley between the two most prominent peaks is characterized regarding its width and depth properties. From these basic features the complete feature set is derived including peak heights, positions as well as the average heights weighted by the respective bin populations. The complete feature set is subsequently reduced by performing a variance analysis to identify the top ten features which contribute the most to the separation of the different cluster-specific frequency distributions.

The following histogram features were identified as the ones having the highest explanatory value regarding the cluster-specific differences of the OFDs:

- Most prominent depth d_1
- 2nd most prominent depth d_2
- Depth of last peak above threshold d_{LLP}
- Prominence of last peak
- Valley width w_{V}
- Sum of trailing values
- Average peak height above threshold
- Average peak prominence above threshold
- Weighted depth average above threshold (green region)
- Weighted depth average of the valley

Feature mapping and correlation analysis

Finally, a set of five features characterizing the keyhole geometries and ten features characterizing their corresponding OFDs have been derived. These feature spaces are the foundation to investigate correlations between real geometric keyhole properties and the OCT signals. Two different measures were applied to quantify these influences. The first is the correlation which is calculated according to Eq. (7) and yields information about the linear dependance between two variables. However, as it is not self-evident that the features are connected to each other in a linear manner, additionally the more general criterion of mutual information (MI) is applied that also takes into account more complex interconnections between two variables (see Eq. (8)).

$$\rho_{i,j} = \frac{\sigma_{i,j}}{\sigma_i \sigma_j} \text{ with } \sigma_{i,j} = \frac{1}{n}\sum_{k=1}^{n}\left(x_k^{(i)} - \overline{x}^{(i)}\right)\left(x_k^{(j)} - \overline{x}^{(j)}\right) \text{ and } \sigma_i := \sigma_{i,i} \tag{7}$$

$$I\left(X^{(i)}, X^{(j)}\right) = \sum_{k=1}^{n}\sum_{l=1}^{n} p_{i,j}\left(x_k^{(i)}, x_l^{(j)}\right) \cdot \log_2\left(\frac{p_{i,j}\left(x_k^{(i)}, x_l^{(j)}\right)}{p_i\left(x_k^{(i)}\right) p_j\left(x_l^{(j)}\right)}\right) \tag{8}$$

In both Eqs. stated above i and j denote the feature indices while n is the sample size and x denotes the value of the respective feature. In Eq. (7), $\sigma_{i,j}$ is the covariance between the features i and j, $\rho_{i,j}$ is the corresponding correlation value and σ_i represents the variance. In Eq. (8), X denotes a feature vector, $p_{i,j}$ represents the joint probability between two feature spaces i and j while p_i and p_j denote the marginal distributions of the respective feature spaces. In praxis, the calculation of the mutual information was achieved by employing an algorithm for estimating the mutual information for discretely distributed variables provided by the Python library *Scikit-learn* (version 1.5.1) (Pedregosa et al., 2011), which is based on the research by Kraskov et al.

Based on a total of ten data points existing in the 5-dimensional feature space of the keyhole geometry analysis and the 10-dimensional feature space of the histographic OCT analysis, the 5×10 correlation matrices are generated per weld experiment. The potential existence of a general correlation between the different feature spaces is obtained by calculating the global interconnections based on the combined features from all experiments. By doing so it is possible to comment on whether a correlation is only pronounced for a specific process with a characteristic process and keyhole dynamics or whether the connection between those feature pairs is of a more general nature.

Results

Cluster analyses of 3D keyhole geometries

The cluster analyses described in “Hierarchical agglomerative clustering” were conducted employing the five geometry features which yielded the least ambiguous results regarding the feature qualification procedure based on labeled artificially generated keyhole models. The scattering of the different keyhole shapes in the 5-dimensional feature space can be visualized using pair plots. The plots on the diagonal show the histograms of the features while the off-diagonal plots show the dispersion of data points with respect to all different pairwise combinations of the features. An exemplary pair plot for the keyhole clustering obtained from a weld carried out at a laser power of 4 kW is shown in Fig. 7.

While some feature combinations like the keyhole front angle and the depth already reveal a separation between certain data point clusters, the more detailed structure remains invisible to the human observer. Applying the cluster analysis to the data yields a dendrogram revealing the number of clusters depending on the feature space distance alongside their affinity. The distance between branches in y-direction indicates the degree of affinity between the two clusters separating at the respective branch. The number of vertical lines intersected by a horizontal line at a certain y-position yields the number of resulting clusters at the given affinity threshold. The dendrogram resulting from the clustering of the data shown in Fig. 7 is given in Fig. 8a).

In addition to the dendrogram of the cluster analysis, the cumulated feature dispersion per cluster alongside the cluster populations are depicted in Fig. 8b). The cumulated feature dispersion serves as a measure of how broadly the members of a cluster a scattered in the feature space and are inversely related to a cluster’s inner cohesion. To evaluate the plausibility of the resulting keyhole clusters a random sample of ten keyhole geometries per cluster is superimposed in a single frame to give an impression of their affinity regarding the human intuition of similarity. As affinity in a strictly mathematical sense can be achieved by an arbitrary set of criteria and there are no objective rules to choose them it is necessary to validate the outcome of the cluster analysis by ensuring agreement between the mathematical affinity and the perceived similarity between members of a cluster. A selection of such superimposed keyhole shapes is displayed in the upper row of Fig. 11. While some clusters exhibit a strong similarity like for example the clusters 1 and 3, others show a broader variety of shapes indicating that data points in that cluster are scattered more broadly. This scattering behavior of the superimposed cluster samples qualitatively matches the cumulated feature dispersions from Fig. 8b). Cluster 1 showing the best sample overlap in the superimposed representation yields the lowest feature dispersion for the welding process at 4 kW while cluster 4 exhibits strong deviations that are in good agreement with the maximum feature dispersion to this welding experiment.

Reproducibility and generalizability of keyhole archetypes

The presented cluster analysis was applied for all weld seams at three different laser powers resulting in a set of ten clusters for each individual weld seam. As mentioned in the introduction, the question arises whether the clusters obtained from different weld seams with the same set of parameters are equivalent to one another and furthermore,

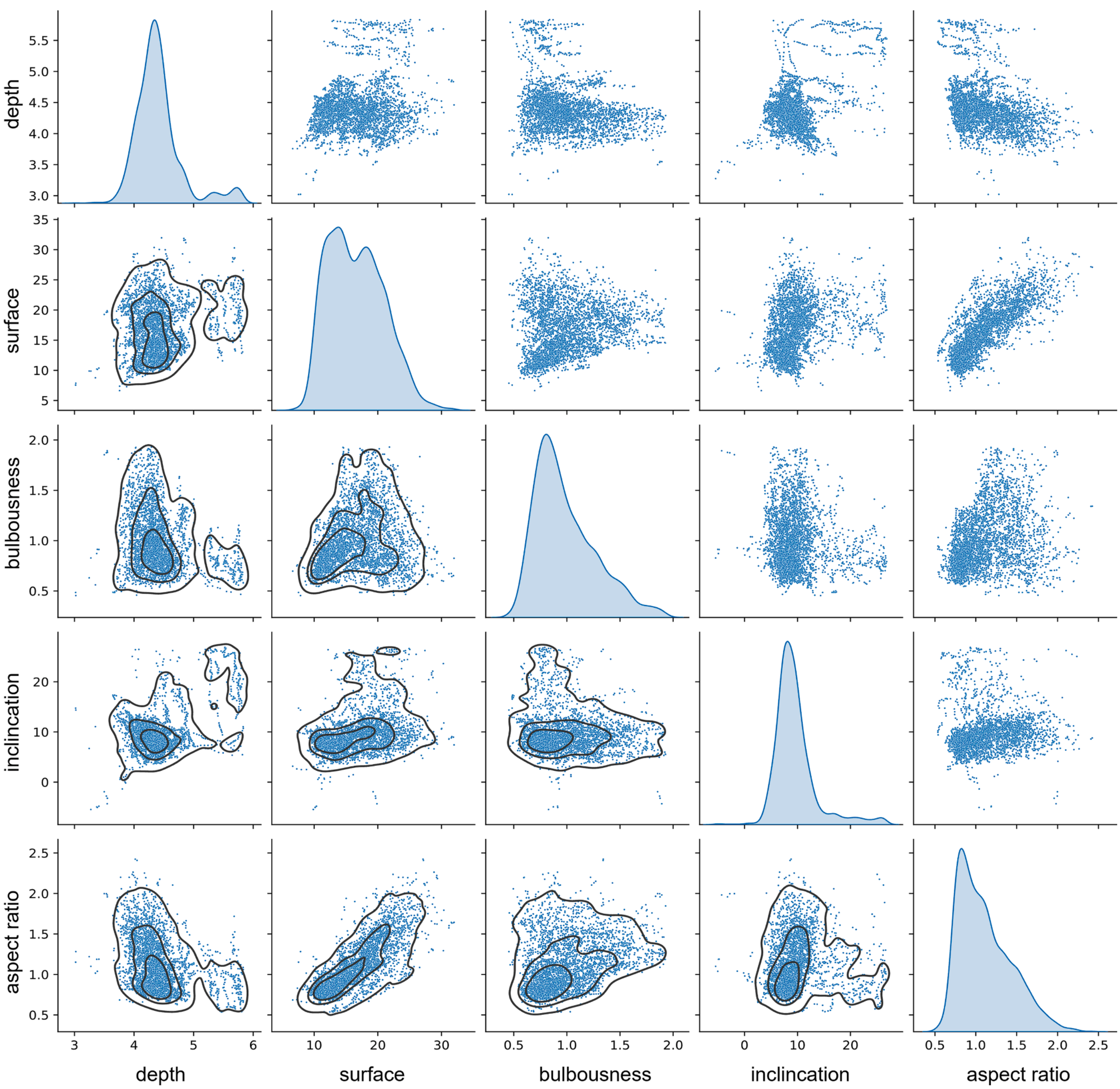

Fig. 7. Pair plots of the keyhole shape representations according to the geometry features for a welding process at a laser power of 4 kW.

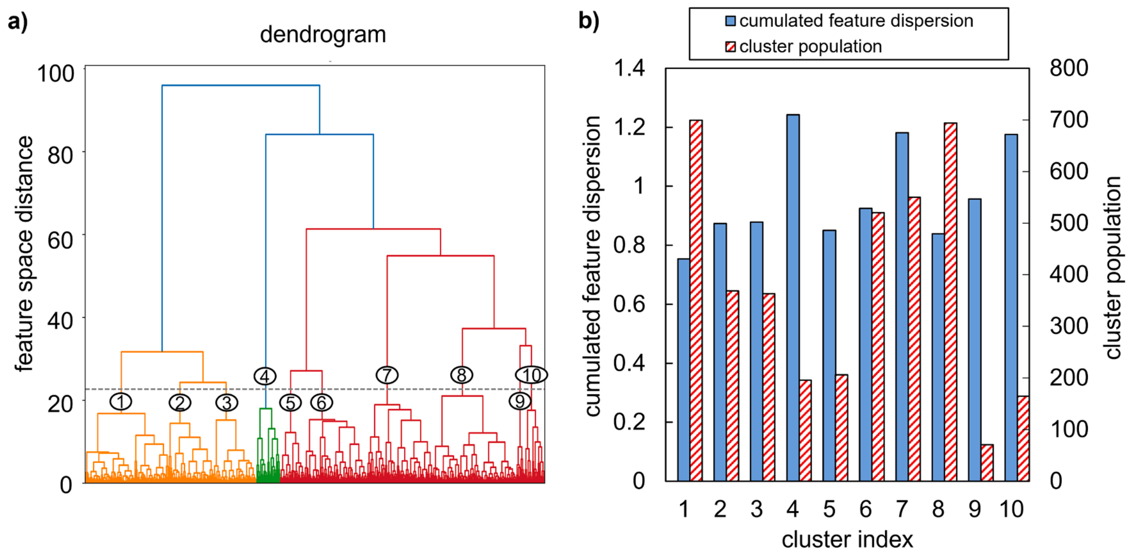

Fig. 8. a) Dendrogram of the hierarchical cluster analysis of keyhole shapes obtained from a welding process at a laser power of 4 kW; b) Column chart of the cluster populations and the cumulated feature dispersion within each cluster (right).

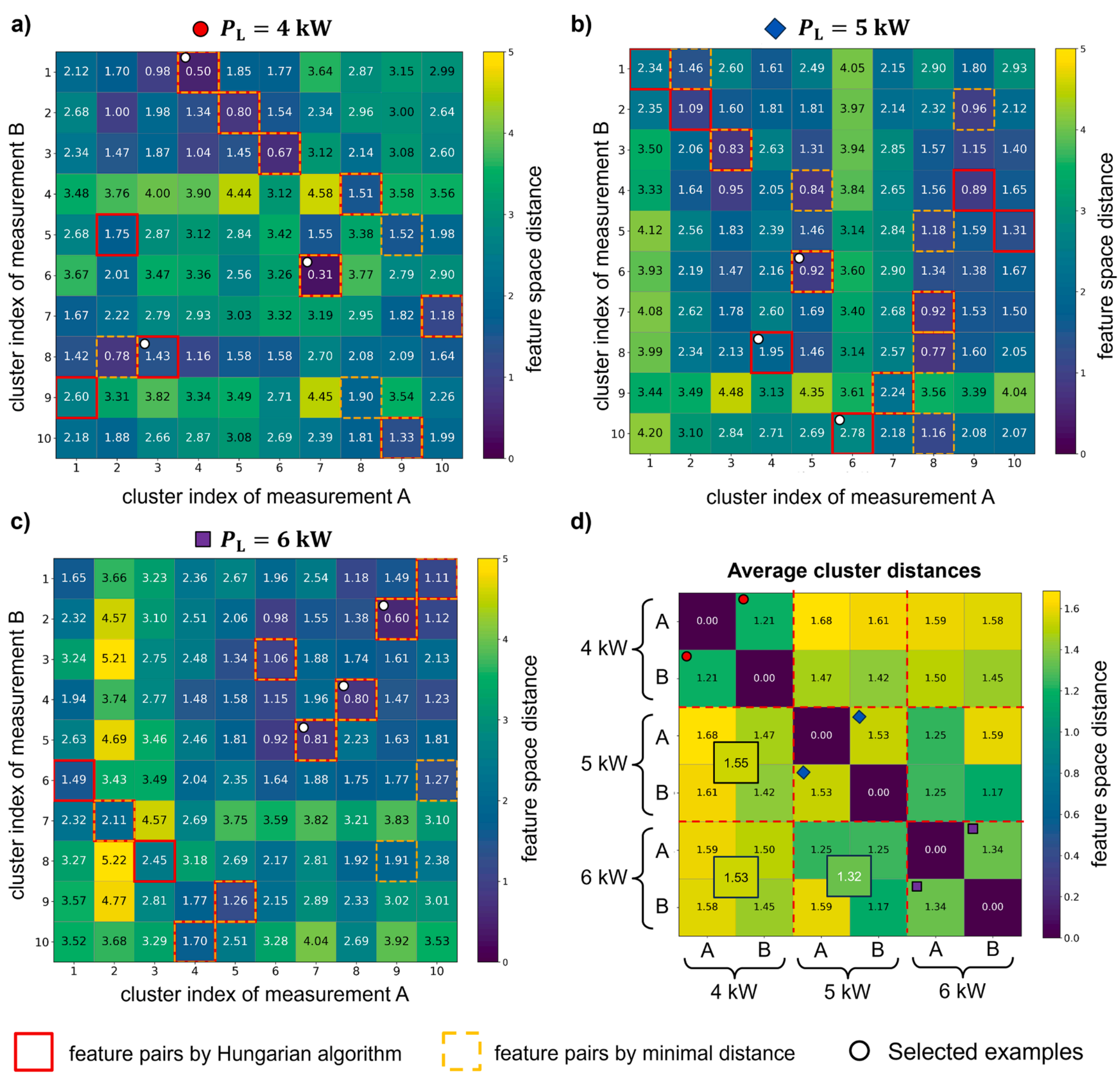

Fig. 9. Cluster distance matrices showing the similarity between clusters obtained from different welds carried out with the same set of process parameters (a – c) and the average cluster distances between all experiments (d).

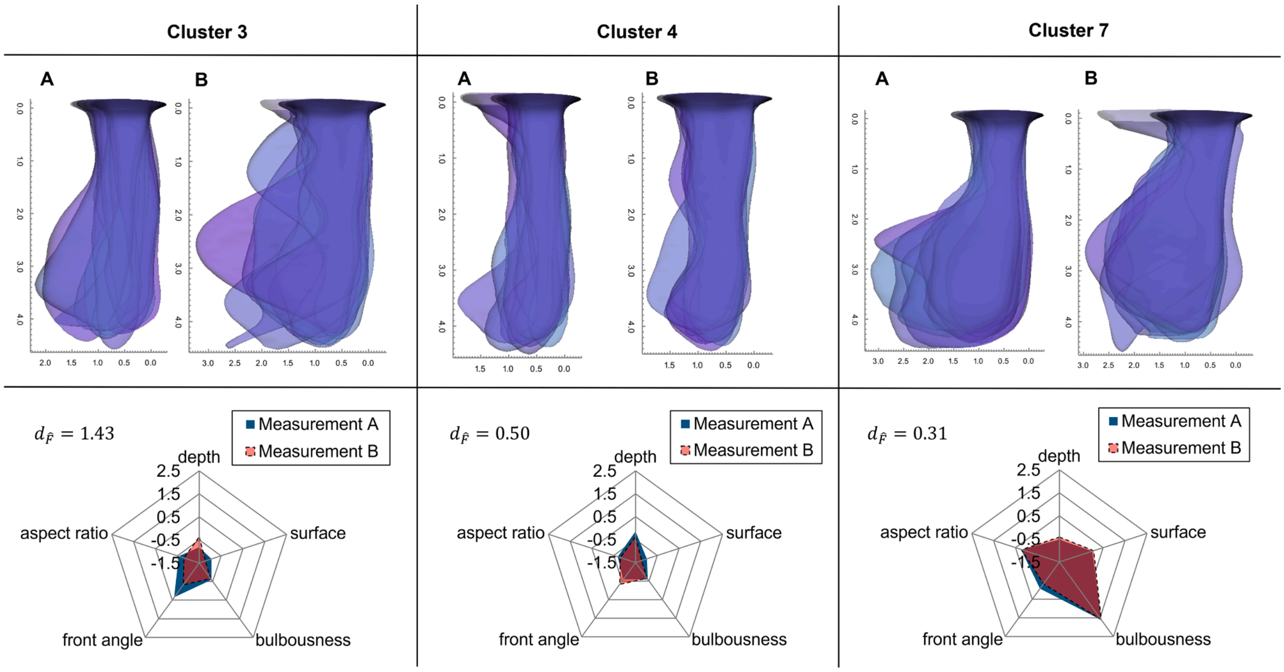

Fig. 10. Examples of equivalent cluster pairs based on the Hungarian algorithm obtained from different welding experiments A and B conducted with the same welding parameters for a laser power 4 kW.

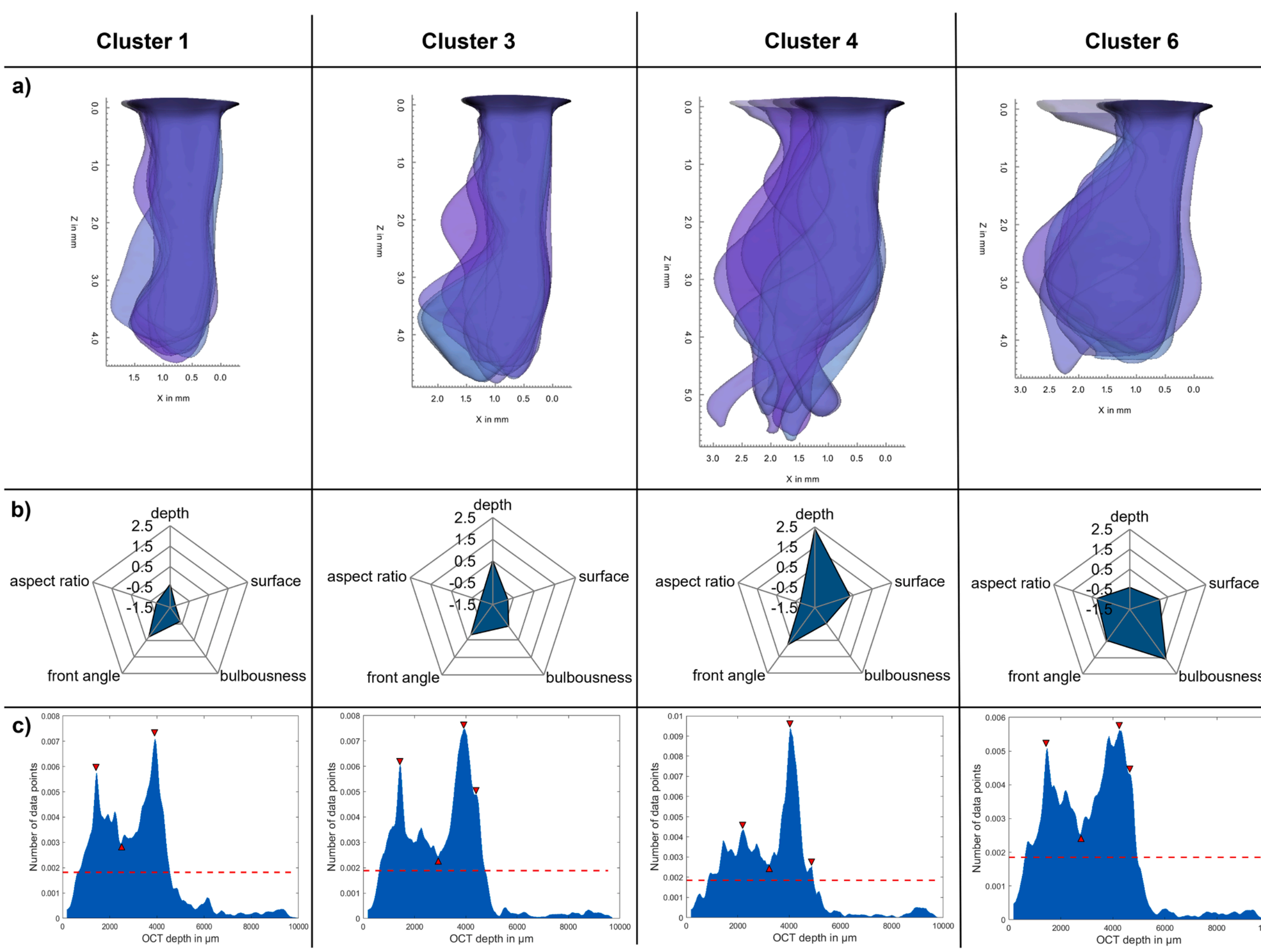

Fig. 11. Exemplary results of cluster analysis and the corresponding OCT frequency distributions for a welding process carried out at a laser power of 4 kW: a) Superimposed keyhole samples for each cluster (sideview); b) Radar charts of the cluster-specific standardized geometry features; c) Corresponding histograms showing the OCT frequency distributions per cluster.

whether there are equivalences between clusters obtained from weld seams at different laser powers. For this purpose, potential cluster equivalencies were investigated according to “Hierarchical agglomerative clustering”. The results are shown in Fig. 9a) to c) where the distance matrices show the pairwise comparison of clusters between different repetitions of welding experiments at the same laser powers. The lower the pairwise feature space distance between clusters the greater the similarity between them. To be able to compare the results the distances are calculated on the basis of the standardized features to focus on relative geometric properties and thus the abstract shapes rather than distinguishing between absolute feature values. The resulting cluster pairs are marked by colored edges where the red edges correspond to the optimal pairs regarding the minimization of the global pair distance according to the Hungarian algorithm while the dashed orange edges correspond to the pairs with the minimum distance. Ideally, both categories of pairs would overlap as indicated by the dashed red and orange edges.

A complete equivalence between two clusters would be indicated by a feature space distance of 0. For all three laser powers comparing the clusters derived from the two repetitions A and B for each weld experiment it can be seen that the cluster distances range between values of 1 and 6. The pairs formed according to the Hungarian algorithm exhibit distances between 0.31 in the best and 2.78 in the worst case indicating that even though the experiments were executed with the same parameters not all keyhole archetypes derived from the HAC are equivalent. The averaged cluster distances between all clusters of different experiments are shown in the matrix in Fig. 9d). The resulting averages of the pairwise cluster distances range between 1.17 and 1.68. As the maximum dissimilarity between two clusters can reach up to a Euclidean feature space distance of 6 these average cluster distances found between different weld experiments indicate an overall similarity between the archetypical keyhole shapes. A selection of equivalent cluster samples obtained at a laser power of 4 kW is depicted in Fig. 10. From the comparison of the overlapping cluster samples it is obvious that a greater perceived similarity is correlated with a significantly lower feature space distance. For cluster 3 the greatest dissimilarity is observed resulting in the highest normalized feature space distance of $d_{\hat{F}} = 1.43$ whereas cluster 7 yields the lowest value of $d_{\hat{F}} = 0.31$ exhibiting the greatest similarity between the randomly drawn cluster samples. The radar charts shown in the bottom row of Fig. 10 support these observations as more similar clusters are characterized by a greater overlap between the covered areas in feature space.

A selection of three clusters each for the remaining laser powers of 5 kW and 6 kW are shown in Figs. 16 and 17 in the appendix and show similar results.

Histographic analysis and feature extraction of cluster-specific OCT signals

According to the results of the cluster analyses, the OCT data points are collected from those time stamps where a keyhole of a certain archetype was present and analyzed histographically as described in “Histographic analysis of the OCT signal”. The resulting histograms for a selected subset of clusters are depicted exemplary in row c) of Fig. 11 along with the superimpositions of their corresponding keyhole clusters

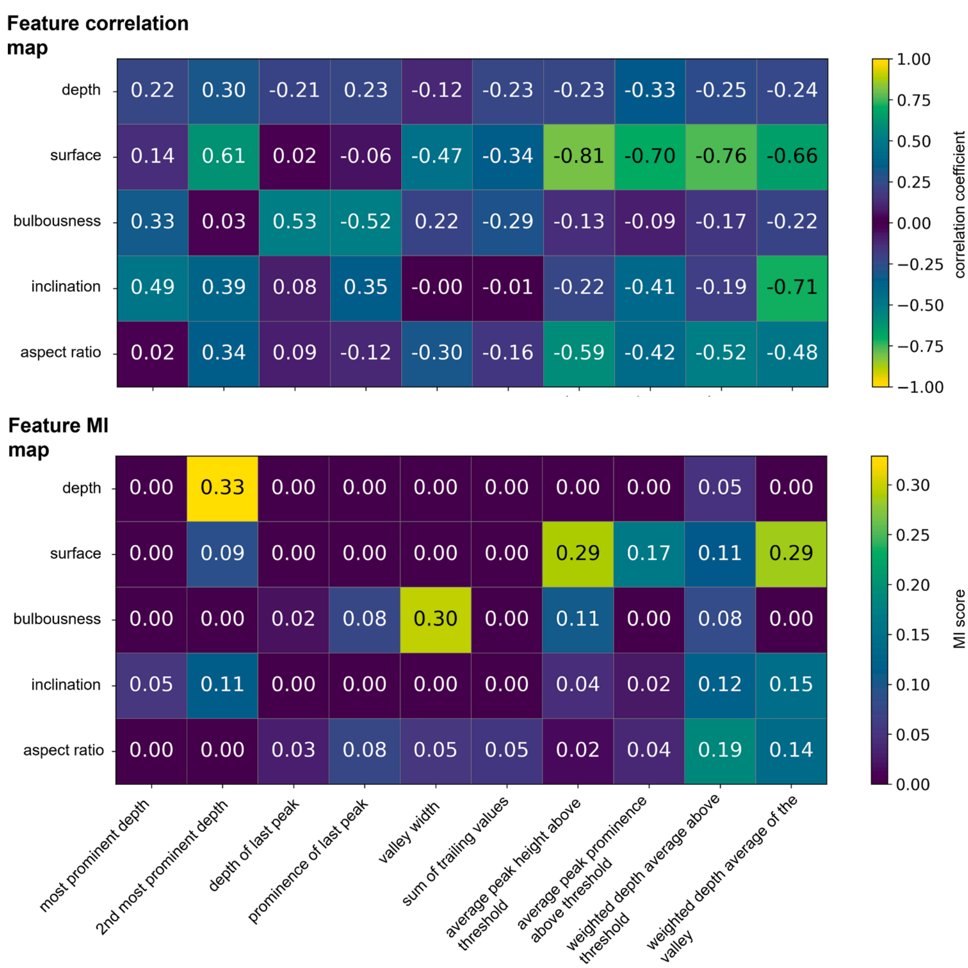

Fig. 12. Feature mapping between keyhole geometry features and OCT histogram features for a welding process carried out at a laser power of 4 kW by means of correlation analysis (top) and MI analysis (bottom).

in row a). The histographic distributions differ qualitatively, indicating the influence of different keyhole types on the OCT measurements. These qualitative differences are quantified by applying the ten previously introduced OFD features. Thus, information contained in the histograms is reduced to a finite set of ten scalar features. For the sake of clarity only the exemplary results for four different clusters obtained during a welding process at a laser power of 4 kW are presented.

From the superimposed side views of the cluster-specific geometry samples in Fig. 11a) as well as from the according radar charts in Fig. 11b) it can be seen that cluster 1 and 3 exhibit great similarities. The shapes are characterized by straight front walls and moderate bulging at the rear wall while the keyholes belonging to cluster 3 are slightly deeper. In contrast, cluster 4 shows stronger front wall inclinations as well as a very narrow tip. Additionally, the superimposed cluster samples exhibit lesser homogeneity which is in good agreement with the cumulated feature dispersion from Fig. 8b) where the dispersion takes on the maximum value for cluster 4. Finally, cluster 6 shows a keyhole archetype with a strongly pronounced bulge resulting in the maximum values for the bulbousness as indicated by the radar chart right below. Accordingly, it is expected that the corresponding histograms for clusters 1 and 3 also show great similarities while for cluster 4 and 6 characteristic differences should become visible. For the presented cluster examples these expectations hold true as the histograms for cluster 1 and 3 are characterized by two pronounced peaks, one in the shallower region and one in the vicinity of the keyhole depth. The resulting valley between these peaks is also quite similar despite being deeper for cluster 3. Also, the last local peak for cluster 3 differs from the most prominent peak and is located at its flank. Cluster 4, despite also having a very pronounced peak in the vicinity of the keyhole ground, shows a completely different distribution where the shallow peak is quite small and the valley almost vanishes. This characteristic corresponds well with what is to be expected for highly pronounced keyhole fluctuations yielding a broader dispersion of OCT data points. Lastly, the bulbous keyhole archetype resulting from cluster 6 exhibits a much broader and more irregular peak in the deeper keyhole region indicating a higher ambiguity of the keyhole ground position from the OCT measurement. Yet the shallow peak is similarly pronounced to the ones from cluster 1 and 3 resulting in significant valley between the two most prominent peaks.

Feature mapping and correlation analysis

The aforementioned observations provide strong evidence that indeed the OCT signal seems to carry information about the keyhole shape beyond its mere depth. Based on these insights analyzing the correlations between geometric features of the keyhole archetypes and the features of the OCT histograms according to “Feature mapping and correlation analysis” allows to quantify the interconnection between the keyhole shape and the resulting OCT signals. The results for the welding experiments carried out with a laser power of 4 kW are depicted in Fig. 12.

The matrix in the upper half of Fig. 12 shows the feature correlation map as a measure of pairwise linear connection between the two feature spaces according to Eq. (7) while the matrix in the lower half represents

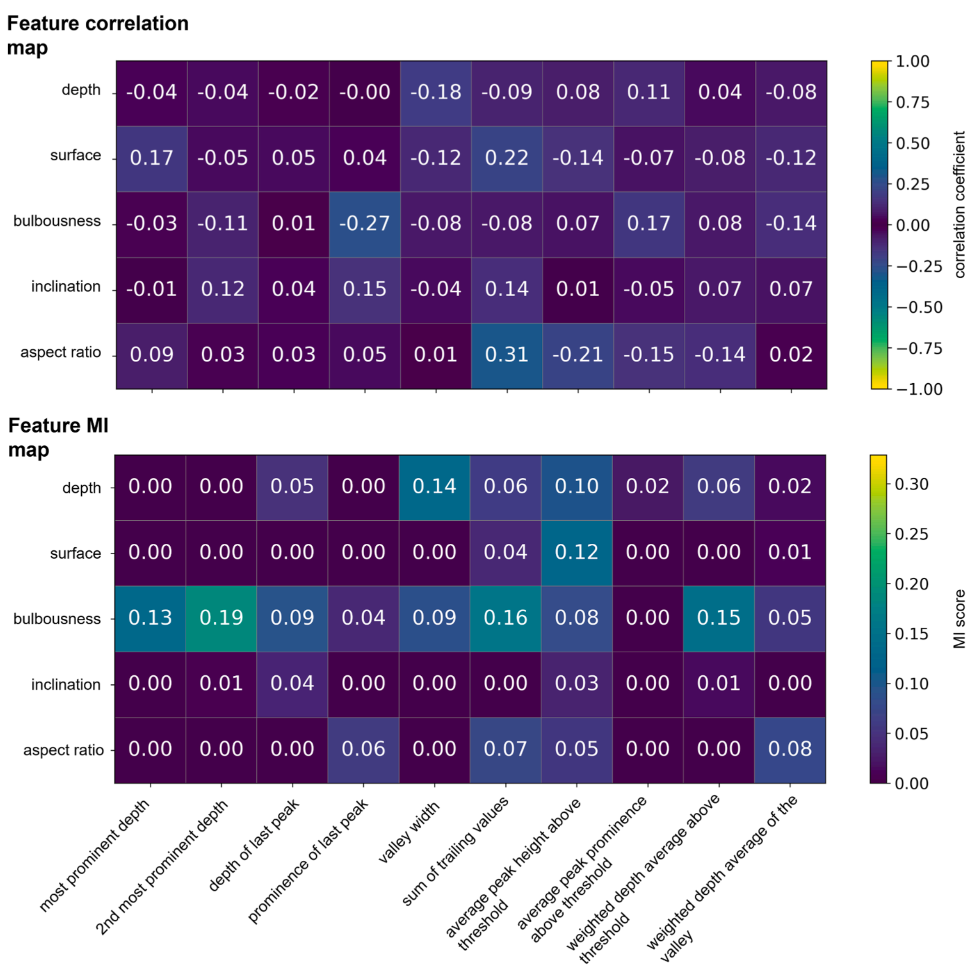

Fig. 13. Global feature mapping between keyhole geometry features and OCT histogram features based on the combined data of all experiments by means of correlation analysis (top) and MI analysis (bottom).

the feature map regarding the mutually shared information according to Eq. (8). From these maps, strong interconnections between features can be identified. The strongest correlation exists between the keyhole surface area and the average peak height above the significance threshold with a negative correlation value of −0.81. Similarly pronounced correlations can be identified between the surface area and the average peak prominence, between the keyhole front wall inclination and the weighted depth average regarding the histogram valley. Furthermore, a positive correlation between the bulbousness of the keyhole and the depth of the last peak as well as a negative correlation between the bulbousness and the prominence of the last peak can be observed both exhibiting absolute values of ~ 0.5. Overall, the keyhole surface area correlates most with the histogram features. The MI analysis accounting for more complex interconnections between the features yields different results. While the identified correlations regarding the keyhole surface area are reproduced by the MI analysis, further interconnections are found between the keyhole depth and the 2nd most prominent OCT depth as well as between the bulbousness and the valley width.

The feature mapping was performed analogously for the experiments at laser powers of 5 kW and 6 kW. The corresponding results are shown in Figs. 18 and 19 in the appendix. At a power of 5 kW the bulbousness emerges as the most relevant keyhole feature in terms of correlation with the corresponding histogram characteristics. In this case four histogram features yield absolute correlation values regarding the bulbousness ≥ 0.82 while all other feature pairs remain below absolute values of 0.5. The bulbousness correlates most strongly with the depth of the last peak, the average peak height and prominence above the significance threshold and negatively with the sum of trailing values. A positive correlation of 0.4 can be observed between the keyhole depth and the depth of the last peak. The MI analysis confirms the importance of the bulbousness feature while additionally identifying interconnections between the keyhole depth with the average peak height above the significance threshold and the corresponding peak prominence which are closely related criteria. Lastly, the results for the welding experiments at a laser power of 6 kW are presented in Fig. 19. Again, the bulbousness emerges as the most important keyhole geometry feature regarding the correlation analysis showing strong correlations with the depths of the highest peaks as well as with the valley width and the sum of trailing values. The highest absolute correlation of −0.99 between the bulbousness and the depth of the last peak contradicts the other experiments and given the unusually high correlation value must be considered an anomaly. Overall, this experiment shows higher correlations so that all keyhole features show correlations with the OCT features of absolute values greater than 0.5. From the MI analysis the interconnection of the keyhole depth with the depth of the last local peak and the weighted depth average of values above the significance threshold as well as the interconnection between the keyhole surface area and the sum of trailing values can be identified as the most pronounced.

Finally, the global feature mapping is performed on the combined dataset of all weld experiments to identify interconnections that are not limited to single experiments and thus have the potential of revealing more general patterns. The results are shown in an equivalent manner to the analyses of the single experiments in Fig. 13. The global correlations are significantly weaker than those for the individual experiments. The correlations range between absolute values from 0.01 to 0.31 where the strongest correlation is observed between the keyhole aspect ratio and the sum of trailing values. Overall, the bulbousness and the keyhole surface area are most prominently correlated to the OFD features. In terms of the MI analyses, the bulbousness emerges as the keyhole features with the most prominent interconnections where the strongest interconnection is given between the bulbousness and the depth of the second most prominent peak as well as the depth of the most prominent peak. Additionally, the keyhole depth exhibits comparatively strong interconnections with the valley width of the OFD as well as with the average peak height above the significance threshold. The strongest interconnections obtained by both measures are in good agreement.

Discussion

The 3D reconstruction of the keyhole geometries is essential for the investigations conducted in this work. For the geometric evolution of the keyhole, a symmetry plane coinciding with the plane defined by the laser beam axis and the traveling direction was assumed. This assumption is justified if the exposure time exceeds the characteristic time scale of process-zone fluctuations. Reported keyhole frequencies range from 1 kHz up to 40 kHz (Nakamura and Ito, 2003) with most frequently stated values in the range between 1.5 kHz and 5 kHz (Schou et al., 1994; Kägeler and Schmidt, 2010) which are comparable to the applied frame rate of the conducted experiments. Consequently, strict lateral symmetry of the process zone cannot be assumed on time average within a single frame. However, the general fluctuation frequencies reported in the literature do not specifically address the lateral keyhole fluctuation. Since keyhole deformations are predominantly observed in the traveling

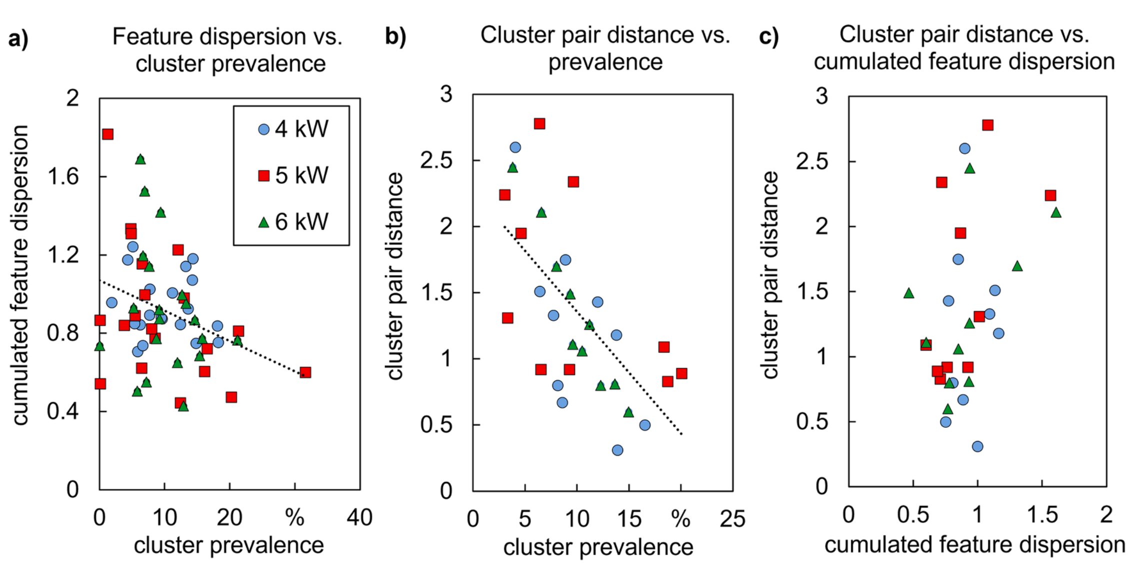

Fig. 14. Analyses of the dependences between cluster properties - a) Dependence of the cumulated feature dispersion of a cluster on its prevalence; b) Dependence of the optimal cluster pair distances from the respective cluster prevalences; c) Connection between cluster pair distances and the cumulated feature dispersion.

direction, manifesting as bulging and constrictions in the trailing region, lateral fluctuations are expected to exert a considerably smaller influence on the propagation of the OCT probe beam. Moreover, potential influences of lateral keyhole fluctuation onto the OCT signal are covered by the statistical evaluation and therefore contribute equally to the presented analyses. The symmetry assumption is therefore not expected to compromise the validity of the reported results.

A total of ten clusters were acquired for each individual weld experiment to achieve a sufficiently high minimal cluster population and at the same time limit the number of clusters to a manageable size. However, the number of clusters can be adjusted for different purposes and different criteria for the cluster merging e.g. inner cohesion could be applied. By inspecting the clustering results the perceivable difference between the clustered keyhole geometries was considered sufficient at ten clusters. It shall be mentioned that for a higher number of clusters which is supposed to yield a higher inner cohesion per cluster, a larger number of X-ray images and thus longer weld experiments would be necessary to ensure a statistically meaningful cluster population. As can be seen from the exemplary superimposition of cluster members in Fig. 10, 16 and 17 the inner cohesion of clusters varies, meaning that the dispersion of the keyhole representations in feature space differs. Therefore, members of some clusters show a greater similarity between each other than those of other clusters. From this observation the question arises whether there is a correlation between a cluster's prevalence on a certain data set measured by the percentage of total contained data points and the inner cohesion expressed in terms of the cumulated feature dispersion. The connection between these properties is depicted in Fig. 14a). The correlation between the two properties appears to be negligible showing only a small negative correlation that is not significant considering the width of the dispersion. Therefore, larger clusters which are considered more representative for the process are not necessarily more cohesive where the inner cohesion is expressed through the cumulated feature dispersion. Thus, more prominent keyhole states do not yield more homogeneous shapes.

The investigation of potential equivalencies between clusters obtained from different experiments conducted with an equal or a dissimilar set of process parameters provides good evidence for the validity of the derived keyhole archetypes in terms of their representativeness of the process. The matrices given in Fig. 9a) to c) show the pairwise feature space distances between cluster centers obtained for different experiments. The feature space distance is given by the Euclidean metrics in the 5-dimensional feature space of the keyhole geometries. The values range from 0.5 to 5.0 for clusters of different experiments conducted with the same set of process parameters. To convey an understanding of those distance values the exemplary cluster comparisons given in Fig. 10, 16 and 17 along with the distance values are presented. Distance values between 0.5 and 1.5 are in good agreement with subjectively perceived similarity while higher values yield increasingly dissimilar pairs. The average optimal cluster differences between different experiments shown in Fig. 9d) are in the range between 1.2 and 1.7 for off-diagonal elements. Equivalent experiments tend to exhibit higher similarities between 1.2 and 1.5 corresponding to a high degree of cluster equivalence between those data sets. However, the strongest similarities of clusters are given between experiments conducted at 5 kW and 6 kW yielding a minimal average cluster distance of $d_{\bar{F}} = 1.17$. Overall, in the given range of process parameters with laser powers between 4 kW and 6 kW the qualitative keyhole shapes are in good agreement. From this observation it can be concluded that the abstractions of the keyhole states by means of HAC are to a high degree representative of the welding process. Therefore, the possibility has been demonstrated that the entirety of keyhole states occurring during a welding process can be systematically and meaningfully reduced by means of unsupervised machine learning techniques.

Based on these insights the question of whether the cluster pair distances correlate with their respective prevalences on a data set is investigated. The underlying assumption states that cluster equivalences are more pronounced the higher the cluster's prevalences are as those clusters are assumed to be more stable and therefore more representative of the respective process. The prevalence of a cluster describes which percentage of the total number of data points in the respective set belongs to it. The correlation between the cluster pair distance and the prevalences is shown in Fig. 14b). Here a clear negative correlation can be observed which is strongest for the data obtained from welding experiments at 6 kW and least pronounced for 5 kW, although for all laser powers the correlation is obvious. These observations support the assumption that bigger clusters yield stronger equivalences between each other due to more stable keyhole states. The correlation between the cluster pair distances and the cumulated feature dispersion or the inner cluster cohesion is shown in Fig. 14c). No correlation between these qualities can be observed leading to the conclusion that more cohesive clusters are not necessarily related to each other more closely.

The histographic frequency analysis of cluster-specific OCT measurements exhibit characteristics that have been captured by the qualified set of histogram features. The primary set of features was selected to provide a certain level of physical interpretability. For example, the histogram peaks are associated with the prevalence of measured surfaces inside the OCT measuring spot while the OCT dispersion is linked to keyhole fluctuations. The distinct interpretation of the features is the goal of this investigation and was approached by means of analyzing interconnections between both feature spaces. The analyses for each experimental parameter set show strong correlations reaching absolute values up to 0.9. Especially the bulbousness criterion, which was designed to express a cavity's potential of entrapping a beam, exhibits the most prominent correlations with the OCT features. This is in good agreement with the expectations as the bulbousness should directly affect the propagation of the OCT measuring beam inside the keyhole. Beside the bulbousness, the keyhole depth and the surface show prominent interconnections with the OFDs although the OCT features vary across different experimental parameters, thus these interconnections do not appear to be universally valid.

The most persistent feature correlations exist between the bulbousness and the depth of the last local peak, the sum of trailing OCT counts and average peak prominence above the significance threshold. The correlation with the depth of the last local peak is negative indicating that the last local peak is merged with more prominent neighboring peaks when the bulbousness increases which could result from a smoothed OCT signal due to multiple reflections inside the keyhole resulting in stronger OCT data point accumulation inside more prominent peaks. The sum of trailing OCT counts is also negatively correlated with the keyhole bulbousness which contradicts the common explanation of high OCT measuring depths exceeding the physically possible depth range by multiple reflections of the probe beam. A pronounced bulbousness should facilitate the occurrence of multiple reflections and thus increase the number of trailing OCT counts in the region of higher depths, however the opposite behavior is observed and instead an overall elevation of the weighted depth inside the valid measuring region is detected.

Finally, the strongest global correlation with the bulbousness is given by the average peak prominence above the significance threshold indicating that peaks in the OFDs are more pronounced for bulbous keyholes. Considering the keyhole depth the strongest correlation occurs with respect to the depth of the most prominent peak of the OFD. As the most prominent peak usually occurs at higher depths close to the last local peak (cf. Fig. 11c)) it is likely to be correlated with the keyhole depth. The correlations with the keyhole surface area are difficult to interpret, especially due to the circumstance that the keyhole width is prone to errors during the geometry reconstruction as noisy X-ray images obscure the measurement of the X-ray extinction.

The evaluation of feature interconnections by means of MI analyses largely confirms the correlations found. However, some additional interconnections are identified which are therefore not linear. Here the most pronounced additional interconnection is found between the keyhole depth and the valley width in the OFD as well as between the bulbousness and the depth of the second most prominent peak which usually corresponds to the shallow peak in the OFD (cf. Fig. 11c)). In general, contrary to the correlation analysis, the MI analysis does not grant access to the qualitative nature of the interconnections between the features but rather makes a statement that some kind of mutually shared information is present in both features. Thus, MI analyses cannot be directly employed for the physical interpretation of the interaction between the OCT probe beam and the keyhole shape but rather indicate where more detailed investigations regarding potential keyhole information encrypted within the OCT signal is advised.

Potentials for future implementation

The presented interconnections between characteristics of the OCT signal and the geometric features of the underlying keyhole shapes emphasize the potential of extracting complex information about the keyhole state and thus the process dynamics solely by means of OCT. Distinct keyhole shapes are known to be indicative of the imminent formation of process pores or keyhole collapses (Engelhardt et al., 2015; Fetzer et al., 2018). Consequently, in-line OCT monitoring could be employed to estimate the probability of defect formation in real time. This would enable targeted and localized post-process quality inspection of weld seams based on defect probability maps.

Beyond defect prediction, insights into the occurrence and stability of certain archetypical keyhole states, as they were proposed in this research, open the perspective of deliberately evoking or stabilizing a certain subset of keyhole states that are considered beneficial for a specific welding application. Such control could be implemented via a feedback mechanism linking the OCT signal to process parameters, e.g. the laser power or the dynamic adjustment of the transverse beam profile. However, for such sophisticated manipulations of the keyhole state further research concerning the stability of keyhole states as well as their controlled adjustment is required.

More generally, the practical relevance of this approach lies in the implicit relationship between the keyhole geometry and the resulting weld quality. Establishing this relationship constitutes an extensive research field in its own right. The central contribution of the present work is therefore the methodological bridge between informationally rich but experimentally costly observations of the keyhole and readily accessible in-line OCT measurements. This bridge is realized through statistically derived mappings between both feature spaces. A comprehensive correlation framework would substantially enhance the capability of OCT-based process monitoring for both industrial quality control and fundamental keyhole research in laser deep penetration welding.

Conclusion

In this study the connection between the OCT signal and the keyhole shape from which it was generated during the deep penetration laser welding of aluminum (EN AW-1050A) was investigated based on simultaneously conducted OCT-measurements coaxially to the processing laser beam and lateral X-ray recordings of the process zone. From the X-ray measurements the keyhole shapes were extracted, and 3D keyhole geometries were reconstructed from the time-resolved gray-scale X-ray images. Based on a set of previously qualified geometric criteria the keyhole geometries were reduced to a finite set of five features capturing the most important qualities which were afterwards subjected to a hierarchical agglomerative clustering analysis to reduce the large variety of keyhole shapes to ten so-called archetypical keyhole shapes. Subsequently, the OCT data points belonging to each keyhole archetype were collected and analyzed statistically by means of a histographic frequency analysis. Again, to characterize the resulting OCT frequency distributions a set of features was derived so that interconnections between characteristic features of the keyhole shapes and their corresponding OCT data point distributions could be investigated. These investigations were performed by means of correlation analyses accounting for linear interconnections as well as by mutual information analyses accounting for more complex feature interconnections. From the results of the investigations presented the following conclusions can be drawn:

- Keyhole shapes can be systematically abstracted to archetypical keyhole shapes representing the most important geometric features of the represented keyhole state.
- Archetypical keyhole shapes are not unique for each welding experiment but exhibit similarities across different welding processes conducted at different process parameters, thus indicating the generalizability of keyhole states.
- The statistical frequency distributions of OCT data points exhibit significant differences depending on which keyhole archetype they were obtained from, thus providing evidence for higher informative content of OCT measurements.
- It was possible to identify strong interconnections between geometric keyhole features and characteristic features of the OCT frequency distributions allowing to retrieve information about the present keyhole shape only from OCT measurements.
- While the most pronounced correlations were observed for specific welding parameters, the global feature mapping across different laser powers also revealed correlations indicating the existence of a more general relation between the keyhole geometry and the according OCT frequency distribution.
- The bulbousness criterion, specifically designed to account for the keyhole's affinity of entrapping incoming beams, was identified as the most influential keyhole feature regarding the characteristics of the OCT frequency distribution.

Based on these findings the initially stated hypothesis that the OCT signal contains more complex information about the keyhole shape beyond the mere keyhole depth could be confirmed. The informative interconnections were identified and quantified.

CRediT authorship contribution statement

Ronald Pordzik: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Eveline Reinheimer:** Writing – review & editing, Software, Resources, Investigation, Formal analysis. **Marcel Möbus:** Writing – review & editing, Investigation. **Thomas Seefeld:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Additional figures

Figs. 15–19

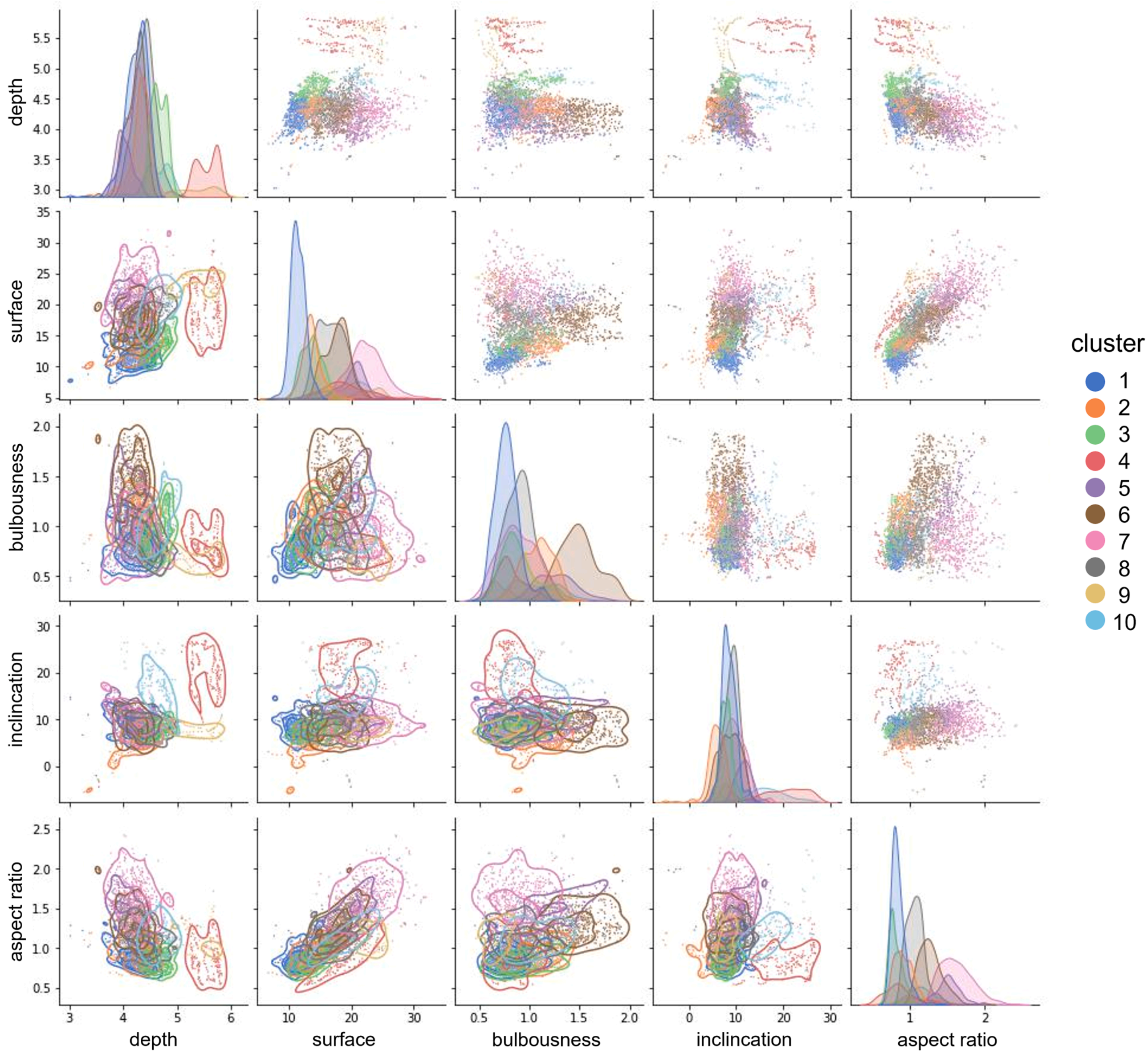

Fig. 15. Pair plots of the clustered keyhole shape representations according to the geometry features for a welding process at a laser power of 4 kW.

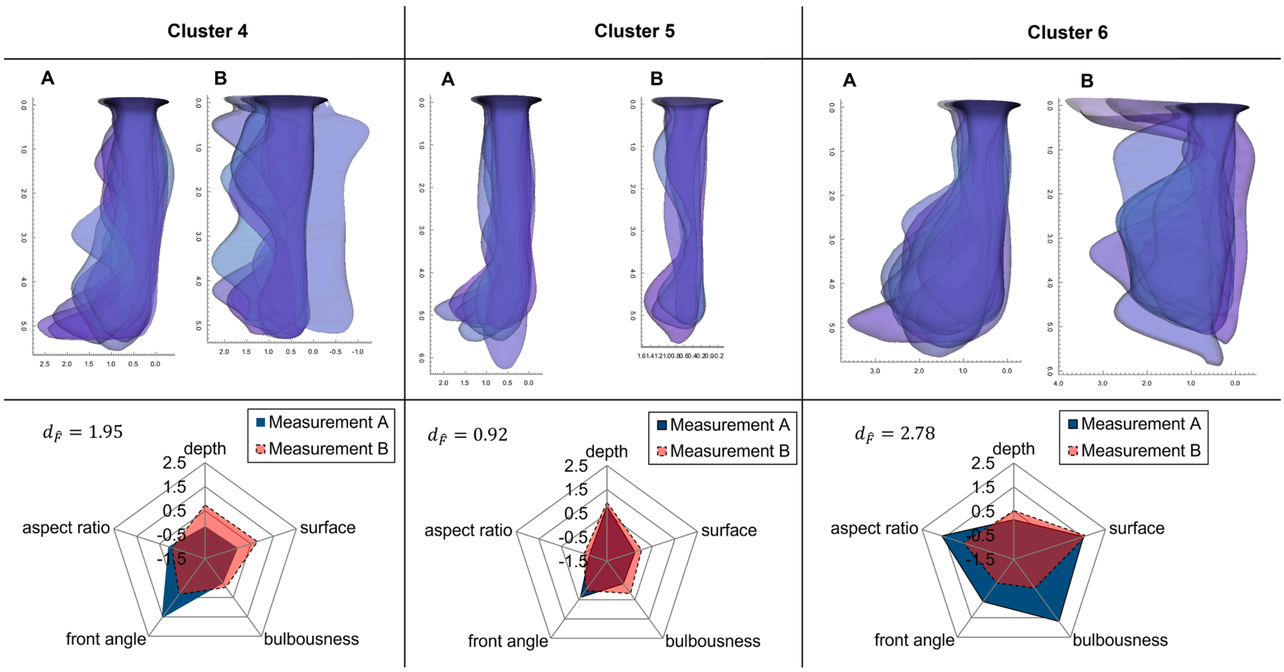

Fig. 16. Examples of equivalent cluster pairs parameters based on the Hungarian algorithm obtained from different welding experiments A and B conducted with the same welding for a laser power 5 kW.

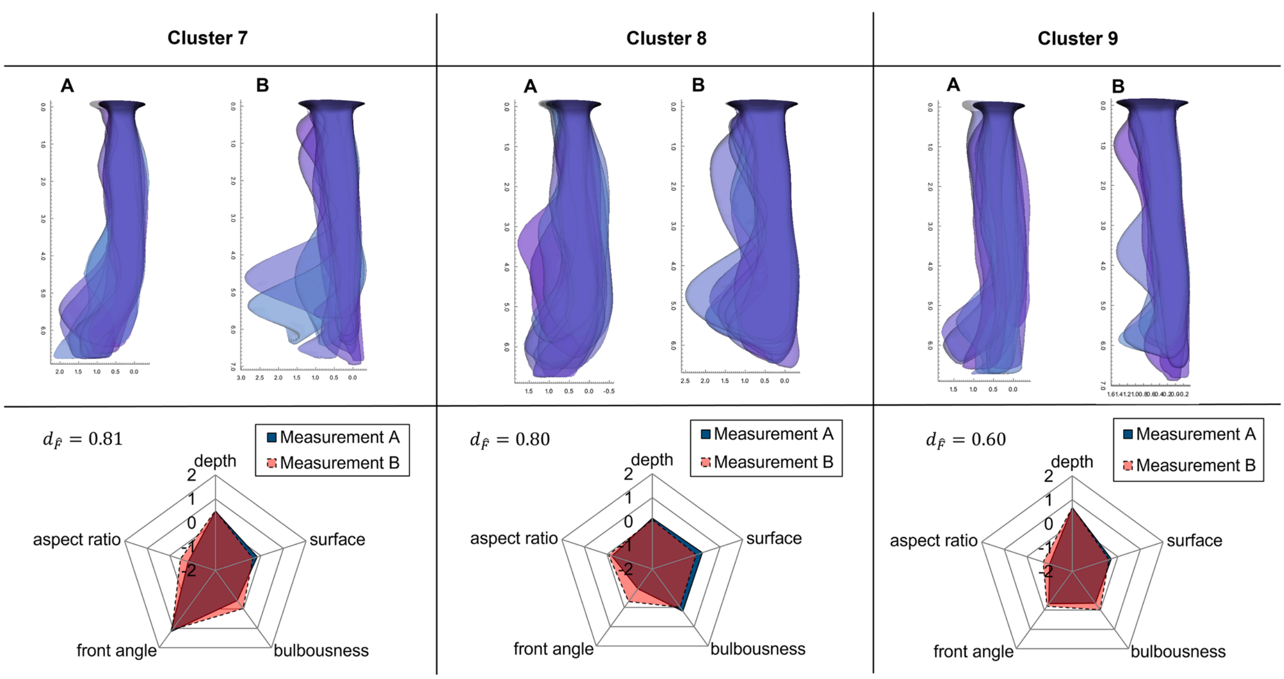

Fig. 17. Examples of equivalent cluster pairs parameters based on the Hungarian algorithm obtained from different welding experiments A and B conducted with the same welding for a laser power 6 kW.

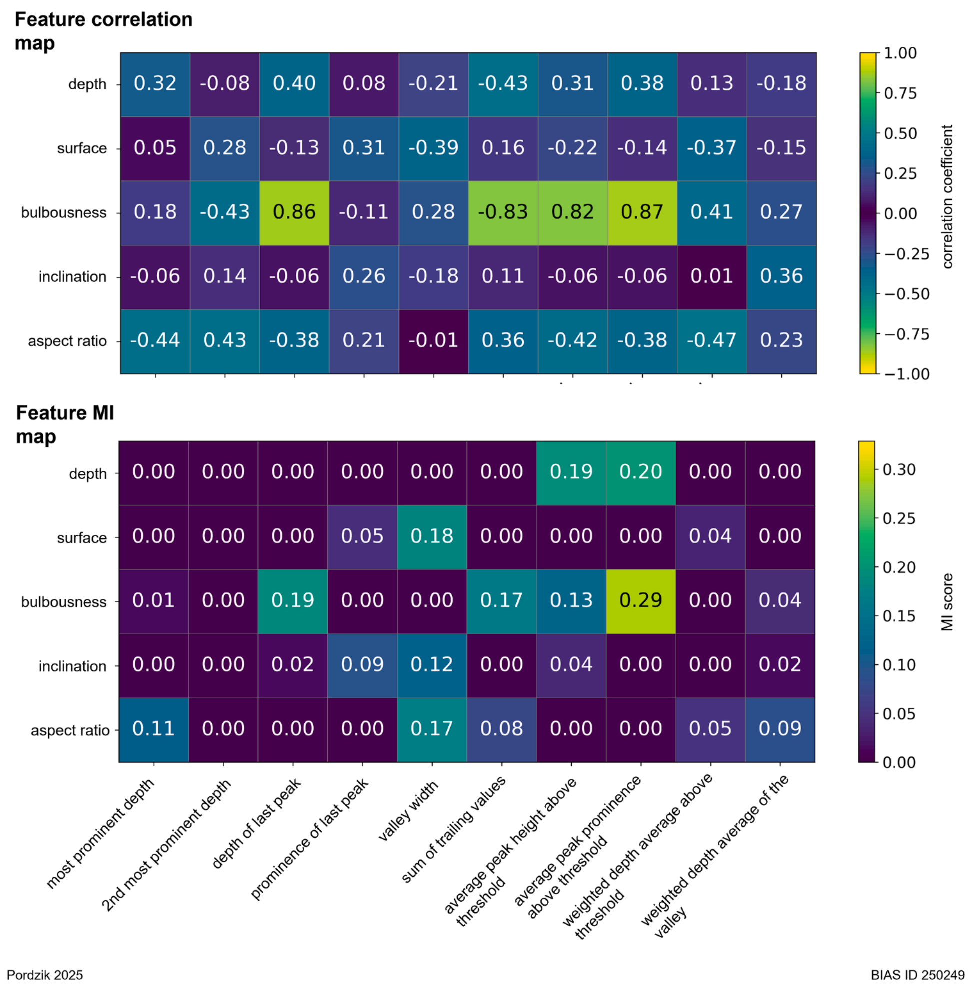

Fig. 18. Feature mapping between keyhole geometry features and OCT histogram features for a welding process carried out at a laser power of 5 kW by means of correlation analysis (top) and MI analysis (bottom).

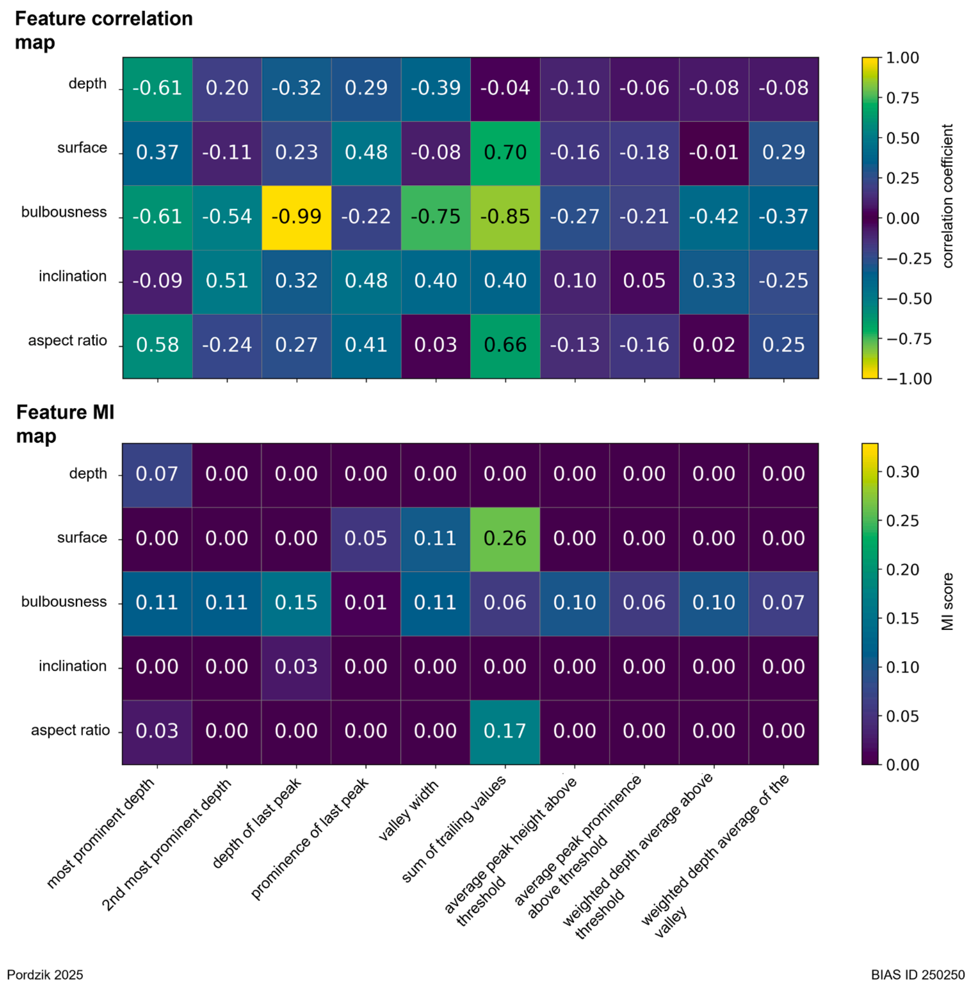

Fig. 19. Feature mapping between keyhole geometry features and OCT histogram features for a welding process carried out at a laser power of 6 kW by means of correlation analysis (top) and MI analysis (bottom).

Data availability

Data will be made available on request.

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